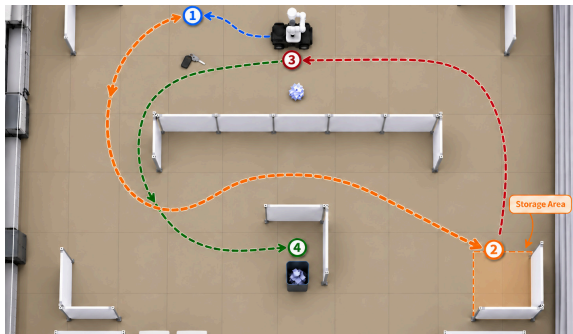


(a) Simple Scenario: Key Storage and Trash Disposal

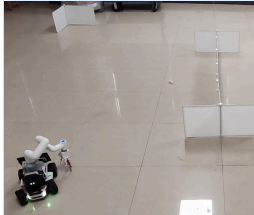
The robot first handles the dropped key, then transitions to the trash-disposal task.

Top-down Map View



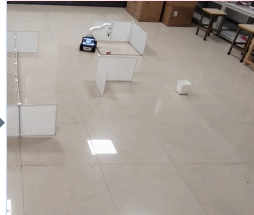
Task Execution Timeline

t1 Grasp key



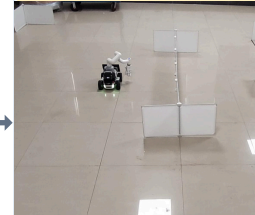
The robot approaches and grasps the dropped key.

t2 Store key



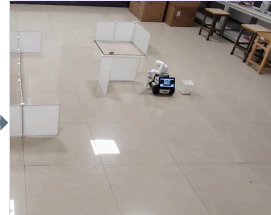
The key is transported to the designated storage area.

t3 Locate trash



The robot transitions to the disposal task and locates the trash.

t4 Dispose trash

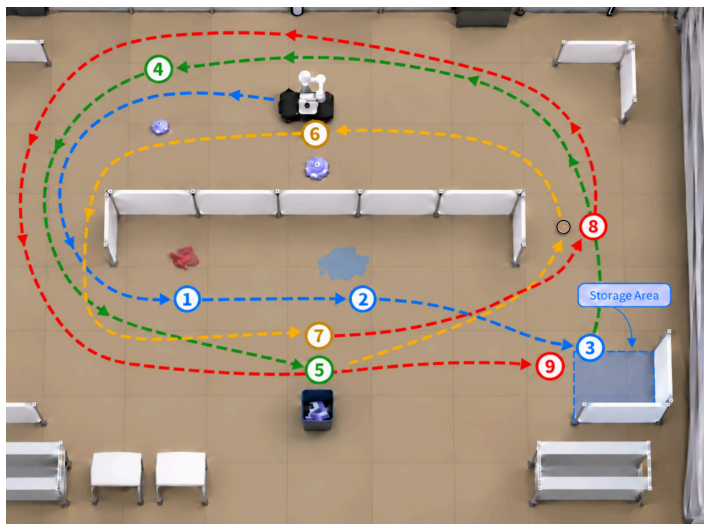


The trash is picked up and delivered to the bin.

(b) Complex Scenario: Semantic Prioritization and Long-Horizon Cleanup

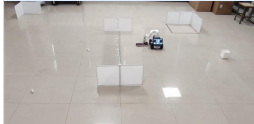
The robot prioritizes the hazardous spill, then sequentially clears the remaining clutter.

Top-down Map View



Task Execution Timeline

t1 Wipe spill



Retrieves the cloth and cleans the spilled water.

t2 Store cloth



Stores the used cloth in the designated storage area.

t3 First trash



Moves to the first piece of trash.

t4 Dispose first



Disposes of the first piece of trash.

t5 Second trash



Moves to the second piece of trash.

t6 Dispose second



Disposes of the second piece of trash.

t7 Pick up tape



Picks up the dropped roll of tape.

t8 Store tape



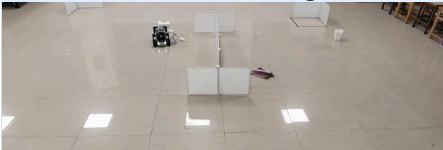
Places the tape in the storage area.

(c) Failure Case: Closed-Loop Corrective Re-execution

Post-action feedback detects the incomplete manipulation task and triggers a retry.

Task Execution Timeline

t1 Initial attempt



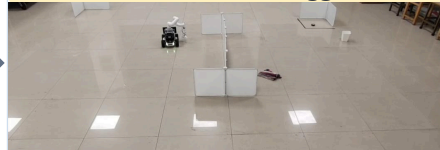
The robot attempts to grasp the second piece of trash.

t2 Manipulation failure



The grasp fails, leaving the disposal task incomplete.

t3 Re-execution triggered



Feedback reveals that the task remains unfulfilled.

t4 Corrective retry



The action is re-issued and the trash is handled successfully.

Fig. X. Real-world hardware validation of CAFE. (a) Simple scenario: the robot stores a dropped key and then disposes of scattered trash. (b) Complex scenario: the robot prioritizes a hazardous spill before executing the remaining cleanup tasks. (c) Failure case: closed-loop feedback detects an incomplete manipulation and triggers corrective re-execution.